Thai Huy Nguyen

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RESEARCH INTERESTS

Open-Endedness **Evolutionary Computation** **Quality-Diversity** **Reinforcement Learning**

EDUCATION

University of Information Technology, Viet Nam National University HCMC *Aug 2024 – Nov 2026 (expected)*
Master of Computer Science
Current GPA: 9.18/10 (completed all course requirements)
Thesis: *Quality and Diversity Optimization for Data Generation with Generative Models*
Advisor: Dr. Ngoc Hoang Luong

University of Information Technology, Viet Nam National University HCMC *Aug 2020 – Jun 2024*
Bachelor of Computer Science
GPA: 9.04/10 *(Top Graduate, Ranked #1 in CS2020)*
Thesis: *Evolutionary Reinforcement Learning for Continuous Control with Stable Phenotypic Variation Operators*
Supervisor: Dr. Ngoc Hoang Luong

RESEARCH EXPERIENCE

University of Information Technology, Viet Nam National University HCMC *Oct 2022 – Present*
Researcher in the Evolutionary Learning and Optimization (ELO) Lab

- Conduct research on Neuroevolution, Quality-Diversity methods for data generation, and Open-Ended Discovery.
- Developed QD methods for adversarial prompt generation in text-to-image and vision-language models.
- Built LLM-assisted QD approaches for natural language generation and automated social-bias discovery.
- Designed phenotypic mutation operators for evolutionary reinforcement learning in continuous control.

SELECTED PUBLICATIONS

[Full Publication List]

Investigating Robustness in Vision-Language Models via Adversarial Prompt Illumination
GECCO 2026 – The Genetic and Evolutionary Computation Conference [paper] [code]
Thai Huy Nguyen, Khoa Tran, Quan Minh Phan, Ngoc Hoang Luong

Diverse and High-Quality Text Generation Assisted by Large Language Models
Knowledge-Based Systems, 2025 [paper] [code]
Thai Huy Nguyen, Ngoc Hoang Luong

Diversifying Adversarial Attacks on Text-to-image Generation
GECCO 2025 Companion – The Genetic and Evolutionary Computation Conference Companion [paper] [code]
Thai Huy Nguyen, Ngoc Hoang Luong

Stable and Sample-Efficient Policy Search for Continuous Control via Hybridizing Phenotypic Evolutionary Algorithm with the Double Actors Regularized Critics
GECCO 2023 – The Genetic and Evolutionary Computation Conference [paper] [code]
Thai Huy Nguyen, Ngoc Hoang Luong

Open-Ended Bias Discovery in Large Language Models
Under review
Thai Huy Nguyen, Ngoc Hoang Luong

SIDE PROJECTS

Integrating Soft Actor-Critic with Descriptor-Conditioned RL MAP-Elites [code]

- Replaced TD3 with SAC in DCRL-ME using QDax library, achieving up to 8× higher QD Score on Humanoid.

JAX Reimplementations of Reinforcement Learning Algorithms [code]

- Successfully reproduced Random Network Distillation in JAX, reaching 6,000 on Montezuma's Revenge after 2B steps.

SCHOLARSHIPS AND AWARDS

Pony Chung Scholarship for Graduate Students ($600) *November 2025*
Pony Chung Foundation; Viet Nam National University HCMC

VinIF Scholarship for Master's Students (∼$4,600) *October 2024*
Institute of Big Data, VinGroup Innovation Foundation - VinIF

Fully Funded Master's Scholarship (∼$2,700) *August 2024*
University of Information Technology, Viet Nam National University HCMC

Merit Scholarship (∼$1,100 total) *2021 – 2023*
University of Information Technology, Viet Nam National University HCMC

SKILLS

Technical Skills	C, C++, Python (PyTorch, JAX, Hugging Face, W&B), Git, Linux, Docker
Languages	English (Fluent), Vietnamese (Native)